

OriginTrace — Oil-Spill Attribution Evidence Package

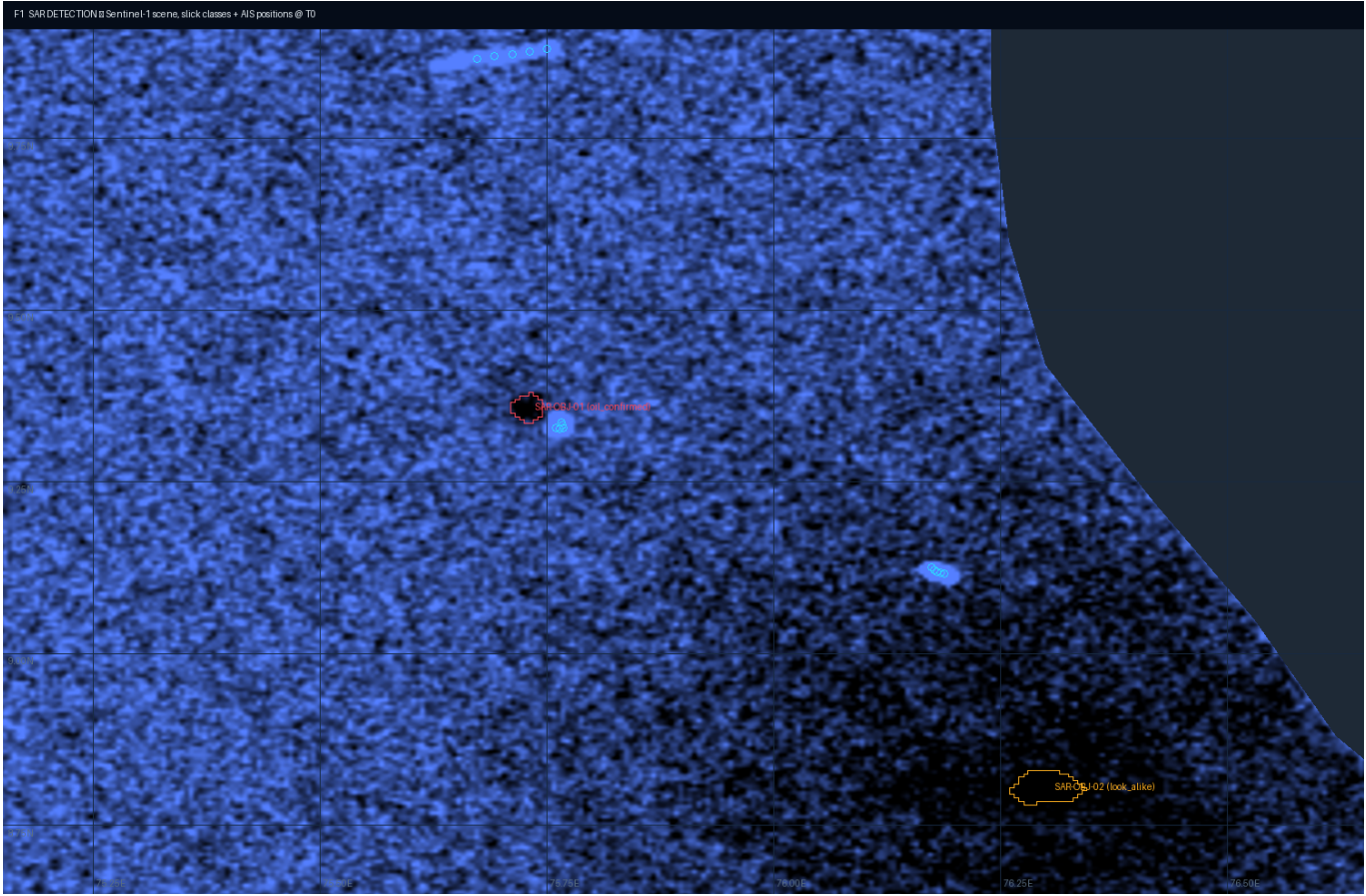
Case: **KERALA_2025_CASE01** | Scene: S1A_IW_GRDH_1SDV_20250612T063000_KERALA_SYNTH | Detection (UTC): **2025-06-12T06:30:00+00:00**
Generated: 2026-08-31 18:01Z | Prepared for: Indian Coast Guard (demo) | SIH-26143 Team Aryan

LEGAL TIER — UNCLOS Art. 220(3): TIP-AND-CUE. This package provides *clear grounds* to request information from the flagged vessels (name, port of registry, cargo log). It is **NOT** detention-grade evidence: Art. 220(5) physical inspection and Art. 220(6) detention require boarding and chemical fingerprinting beyond this system.

1 · Satellite Detection (Sentinel-1 SAR)

Object	Class	Area (km²)	Contrast (dB)	Wind (m/s)	Conf.
SAR-OBJ-01	oil_confirmed	11.0	2.48	7.2	0.77
SAR-OBJ-02	look_alike	29.8	1.86	1.7	0.35

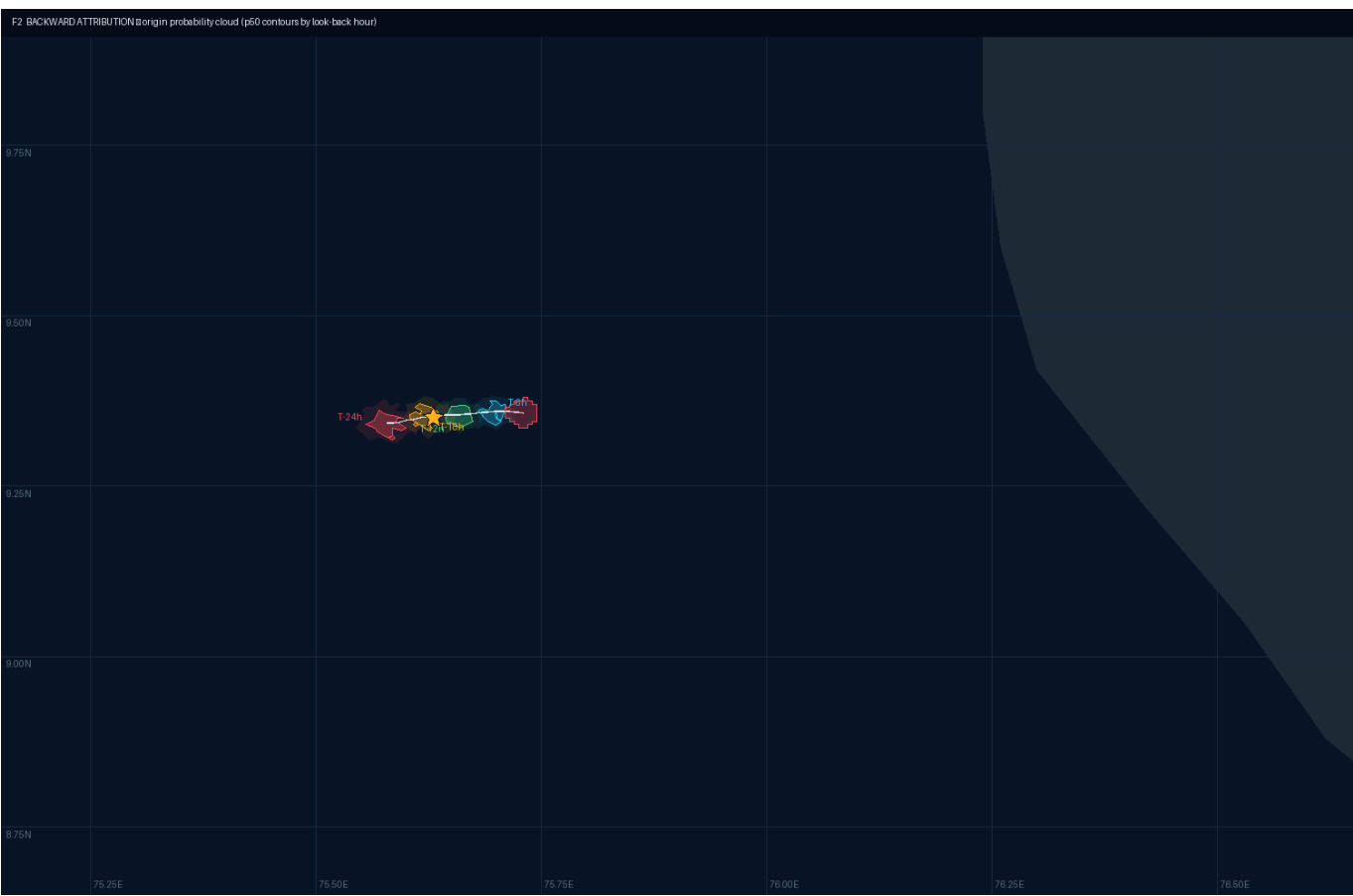
Detection is considered **reliable only in 3–12 m/s wind**; objects in low-wind are typed *look-alike* (see detectability mask on map layer).



2 · Backward Attribution (Origin Reconstruction)

350 Lagrangian particles advected **backward** 24 h from the detected slick (current + $0.03 \times U_{10}$ windage + diffusion, $dt=600$ s). Hourly KDE clouds $\rightarrow$ p10/p50/p90 probability-mass contours.

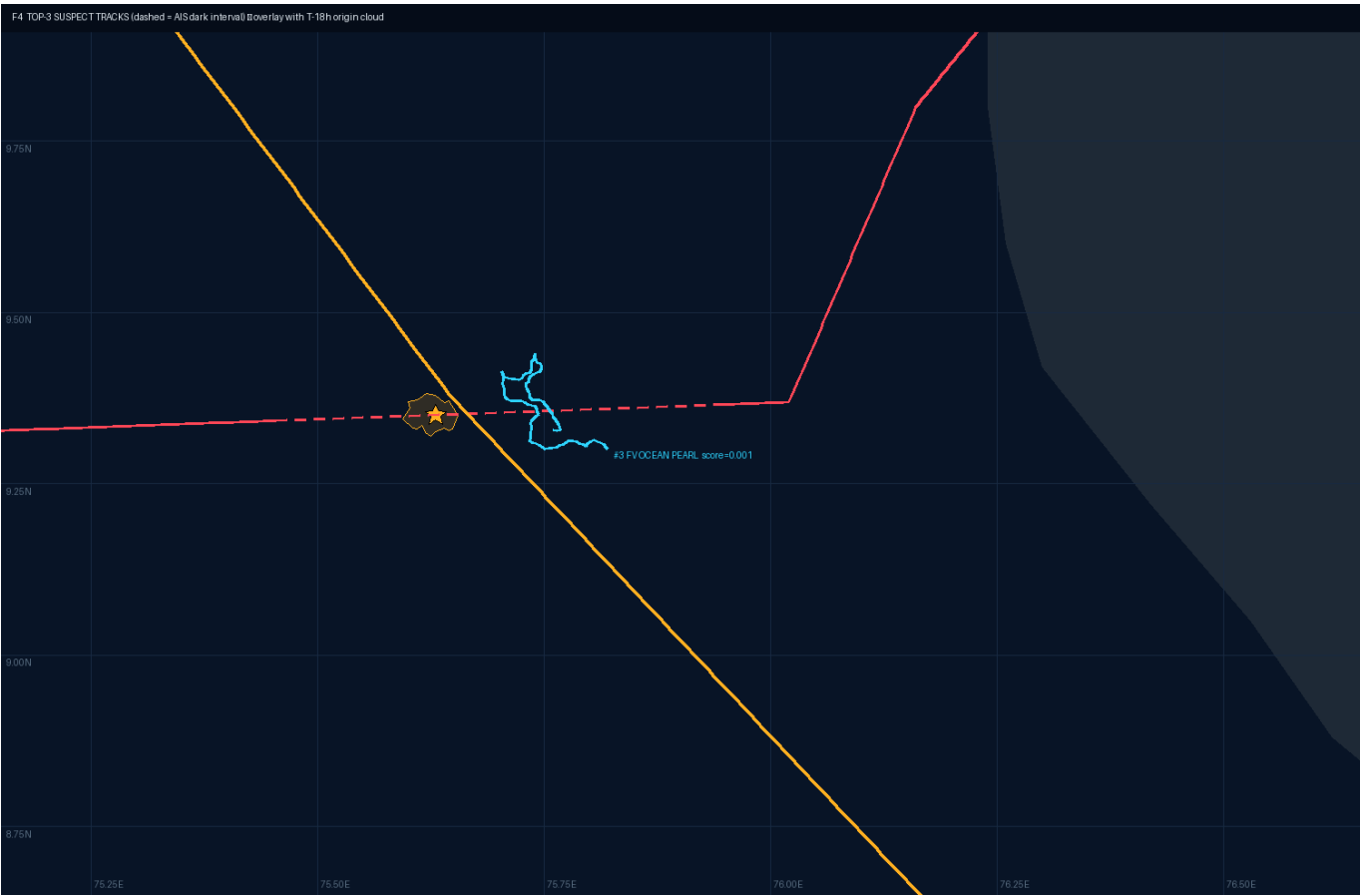
Estimated origin	Release window (UTC)	Engine
75.63E, 9.35N	2025-06-11T08:30Z -> 2025-06-11T16:30Z	OriginTrace Lagrangian Engine v1.0



3 · Vessel Attribution — Ranked Suspects

Model: LightGBM binary classifier (synthetic-calibrated) + TreeSHAP (pred_contrib) explanations. Release window scored: 2025-06-11T06:30Z → 2025-06-12T02:30Z. 20 vessels evaluated.

#	Vessel	MMSI	Verdict	Score	Leading factors
1	MT KAVERI STAR	419000001	PRIMARY SUSPECT	1.000	position coincides with the cloud at release-relevant hours; closest approach 0 km to the reconstructed release locus; vessel went dark for 6.2 h inside the release window
2	MV MALABAR TRADER	419000002	SUSPECT — CORROBORATED	0.790	position coincides with the cloud at release-relevant hours; best 3 h sub-window shows slow, straight-run steaming; closest approach 1 km to the reconstructed release locus
3	FV OCEAN PEARL	419000003	CLEARED	0.001	position coincides with the cloud at release-relevant hours; track intersects the reconstructed origin cloud at plausible times; closest approach 5 km to the reconstructed release locus



4 · Chain of Custody — SHA-256

Artifact	SHA-256
sar_scene (scene.tif)	8aa5c7904dcd2c22e6059b7ae6809ad870a4c8f5f07e9bebfac7cb8840f8e9bc
slick_polygons.geojson	6ad14d8e9d0de45fec5e48b962dd38d430bda660d2fd285664a4bb28e7e6af14
backtrack_hourly.geojson	be0d72918dedb54a239a803e9207757c1eed9eb32e254ce05eb861a426b21ca
suspects.json	00a61db92d178519867a7be2d0965173e849d0ba88b171e6597678e9db6ad38a
lgbm_ranker.txt	473f77f0552b378e688aa0b69e430b53a29cc1e8e92db618f7b417527bcc921c

PACKAGE ID (hash of hashes)	846d3ad85324c3704b864523806a643d10dad3d12542abf940540813b8fd9d71
-----------------------------	--

Synthetic demo case — generated scene/AIS/met-ocean stand-ins per the SIH-26143 build plan (Zenodo-fallback mode). Method chain is identical for operational Sentinel-1 / AIS feeds. OriginTrace is a decision-support system; all findings require corroboration per the UNCLOS enforcement ladder (220(5) inspection, 220(6) detention, MARPOL sampling).